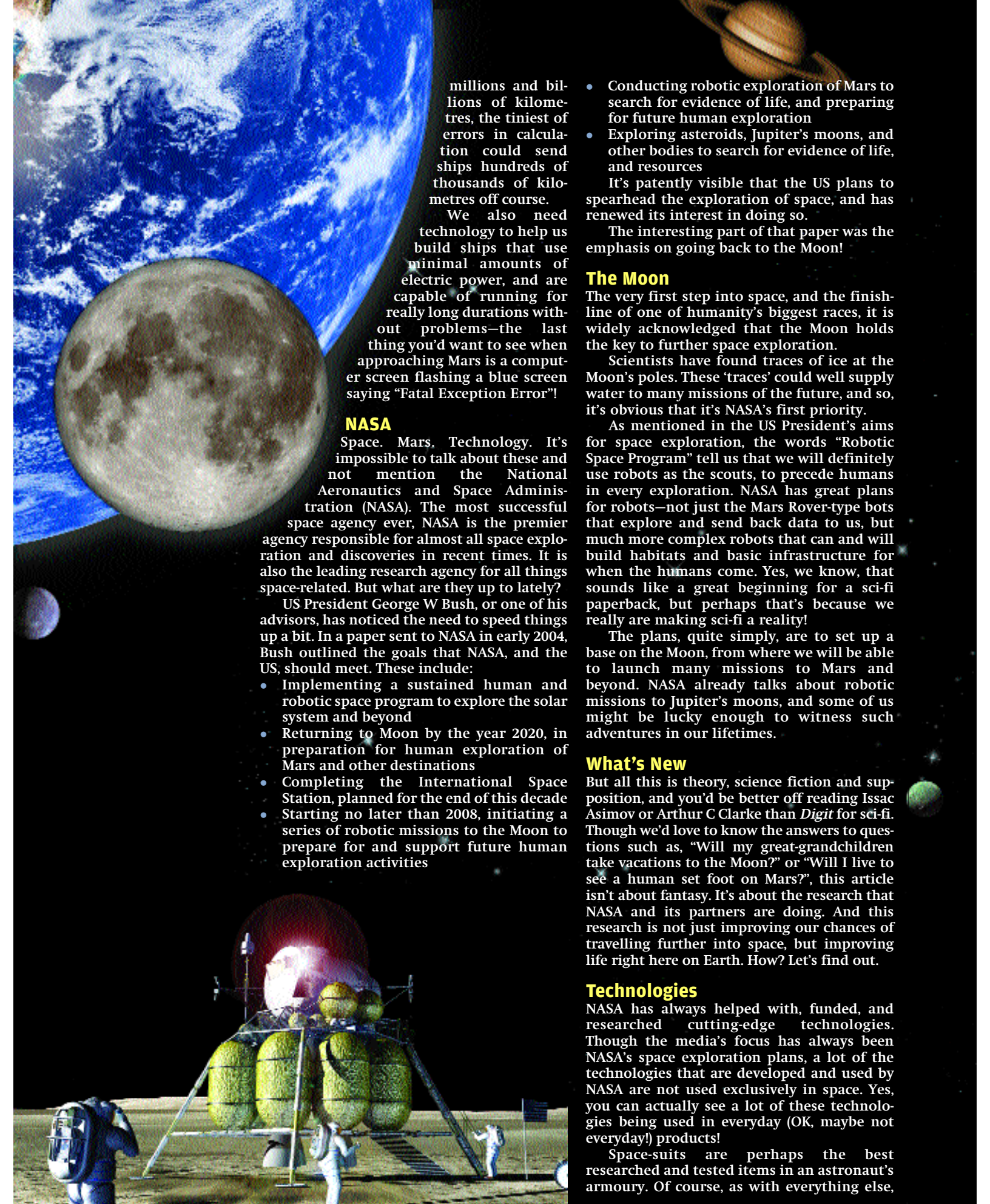




millions and billions of kilometres, the tiniest of errors in calculation could send ships hundreds of thousands of kilometres off course.

We also need technology to help us build ships that use minimal amounts of electric power, and are capable of running for really long durations without problems—the last thing you’d want to see when approaching Mars is a computer screen flashing a blue screen saying “Fatal Exception Error”!

NASA

Space. Mars. Technology. It’s impossible to talk about these and not mention the National Aeronautics and Space Administration (NASA). The most successful space agency ever, NASA is the premier agency responsible for almost all space exploration and discoveries in recent times. It is also the leading research agency for all things space-related. But what are they up to lately?

US President George W Bush, or one of his advisors, has noticed the need to speed things up a bit. In a paper sent to NASA in early 2004, Bush outlined the goals that NASA, and the US, should meet. These include:

- Implementing a sustained human and robotic space program to explore the solar system and beyond
- Returning to Moon by the year 2020, in preparation for human exploration of Mars and other destinations
- Completing the International Space Station, planned for the end of this decade
- Starting no later than 2008, initiating a series of robotic missions to the Moon to prepare for and support future human exploration activities

- Conducting robotic exploration of Mars to search for evidence of life, and preparing for future human exploration
- Exploring asteroids, Jupiter’s moons, and other bodies to search for evidence of life, and resources

It’s patently visible that the US plans to spearhead the exploration of space, and has renewed its interest in doing so.

The interesting part of that paper was the emphasis on going back to the Moon!

The Moon

The very first step into space, and the finish-line of one of humanity’s biggest races, it is widely acknowledged that the Moon holds the key to further space exploration.

Scientists have found traces of ice at the Moon’s poles. These ‘traces’ could well supply water to many missions of the future, and so, it’s obvious that it’s NASA’s first priority.

As mentioned in the US President’s aims for space exploration, the words “Robotic Space Program” tell us that we will definitely use robots as the scouts, to precede humans in every exploration. NASA has great plans for robots—not just the Mars Rover-type bots that explore and send back data to us, but much more complex robots that can and will build habitats and basic infrastructure for when the humans come. Yes, we know, that sounds like a great beginning for a sci-fi paperback, but perhaps that’s because we really are making sci-fi a reality!

The plans, quite simply, are to set up a base on the Moon, from where we will be able to launch many missions to Mars and beyond. NASA already talks about robotic missions to Jupiter’s moons, and some of us might be lucky enough to witness such adventures in our lifetimes.

What’s New

But all this is theory, science fiction and speculation, and you’d be better off reading Issac Asimov or Arthur C Clarke than *Digit* for sci-fi. Though we’d love to know the answers to questions such as, “Will my great-grandchildren take vacations to the Moon?” or “Will I live to see a human set foot on Mars?”, this article isn’t about fantasy. It’s about the research that NASA and its partners are doing. And this research is not just improving our chances of travelling further into space, but improving life right here on Earth. How? Let’s find out.

Technologies

NASA has always helped with, funded, and researched cutting-edge technologies. Though the media’s focus has always been NASA’s space exploration plans, a lot of the technologies that are developed and used by NASA are not used exclusively in space. Yes, you can actually see a lot of these technologies being used in everyday (OK, maybe not everyday!) products!

Space-suits are perhaps the best researched and tested items in an astronaut’s armoury. Of course, as with everything else,